清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

أرامكو السعودية
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Assessment of Iron and Steel Technologies from A Global Perspective

Tsinghua University & Saudi Aramco

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清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

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Project Timeline

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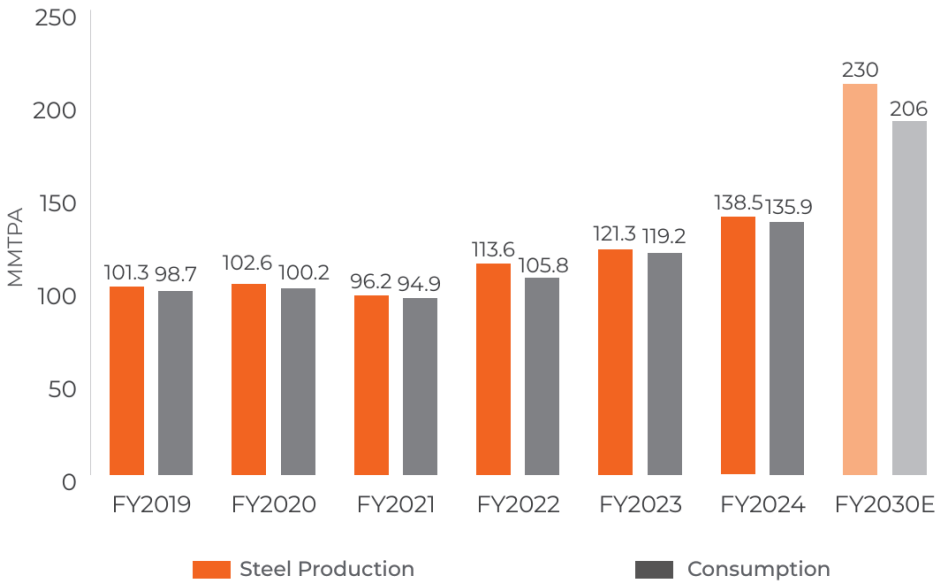
	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July
Research Proposal & preparations																								
Overview of global steel trade & technology assessment																								
Data requirement & database investigation (Base on case India)																								
Model development (Base on case India)																								
Case study: India																								
Case study: USA & Saudi Arabia																								
Case study: Australia & Japan																								
Case study: Germany & Russia																								
Case study: China-Australia steel trade route																								
Final report																								

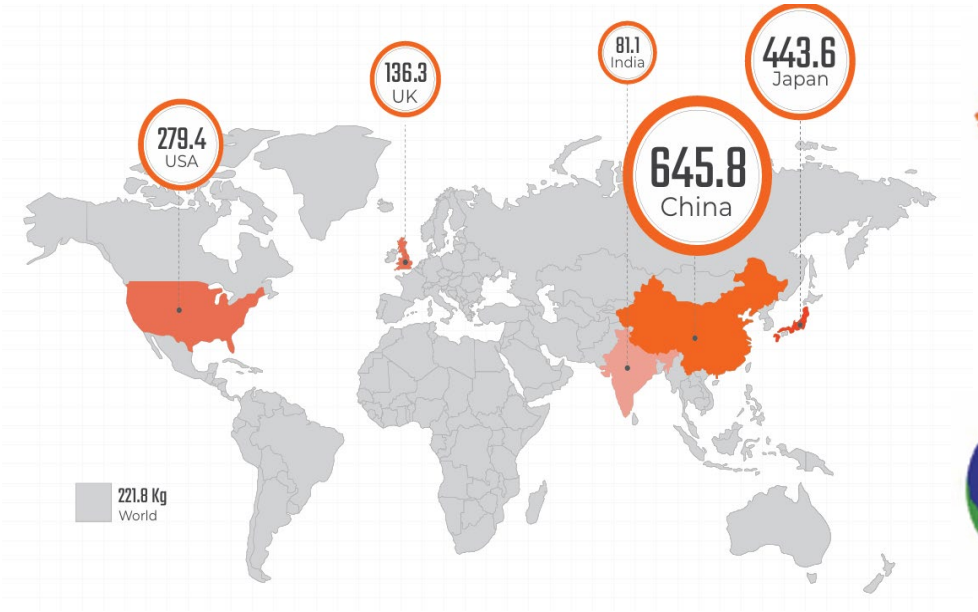
Note1: Models and data requirements lead to each other's updates, so we often perform these two processes at the same time.

Note2: As models and data are continuously updated, case studies will be needed to assess whether they're effective - so we use the Indian case as an example.

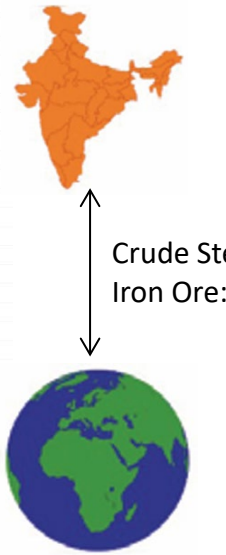
India's steel industry has huge development potential



India's steel production and consumption volume in recent years



Comparison of Annual Per Capita Steel Consumption: India vs. Key Countries

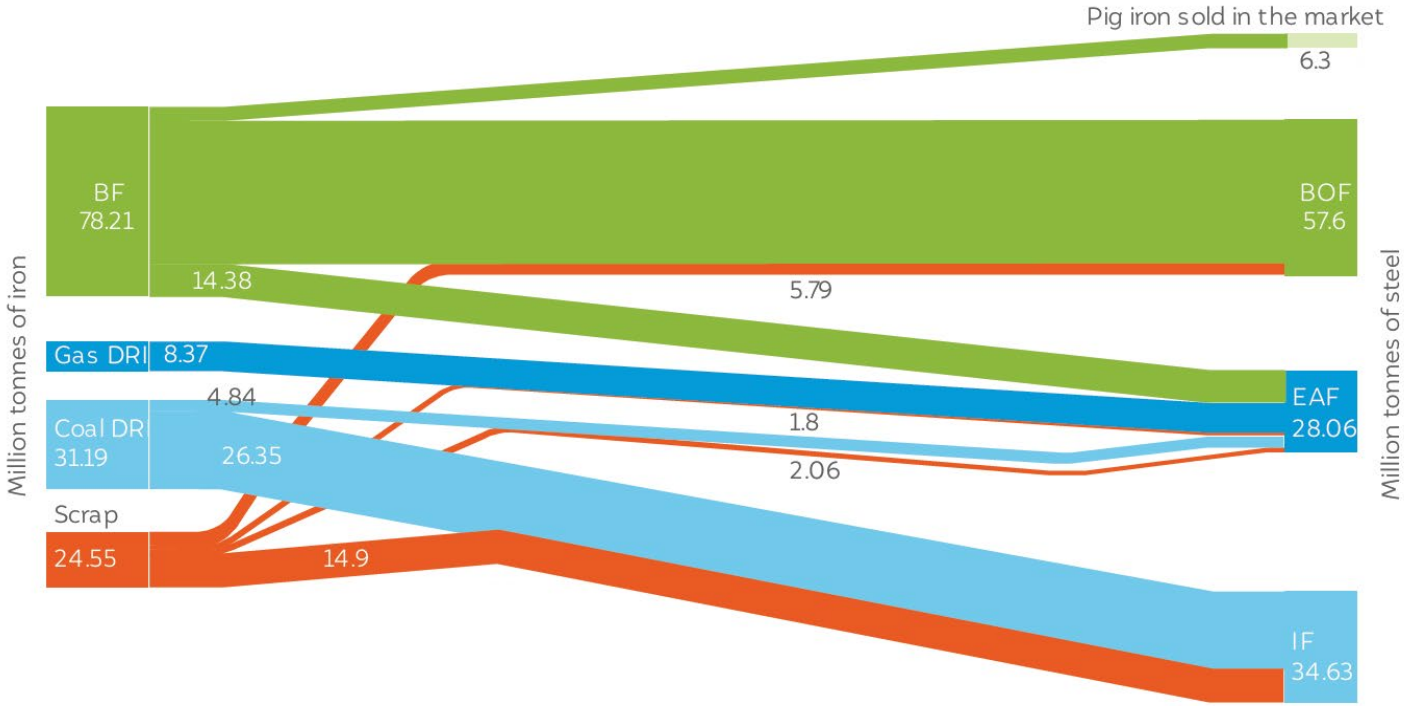


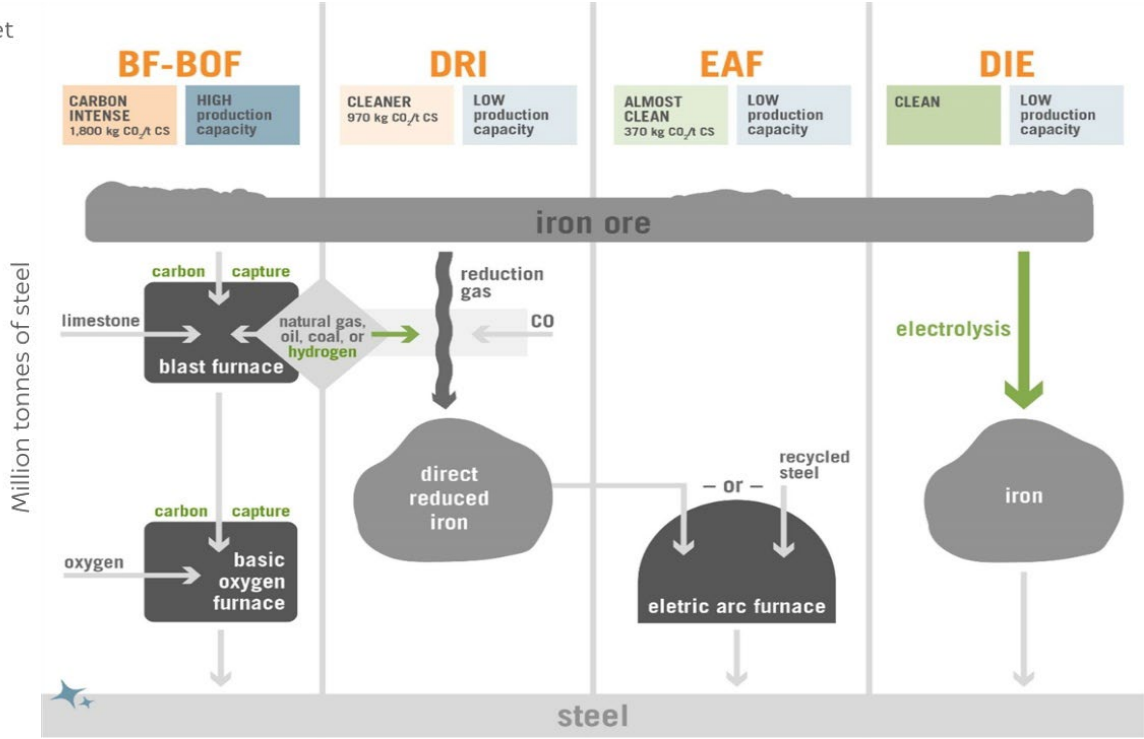
Crude Steel: Net Import of 2.3Mt
Iron Ore: Net Export of 14 Mt

Import & output for Indian steel industry

- India's steel consumption & production's growth is expected to continue.
- The vast majority of India's steel needs can be met through domestic production.
- India's iron ore production can meet its consumption needs, and even export a certain amount of iron ore.

Main technical routes of the India steel industry



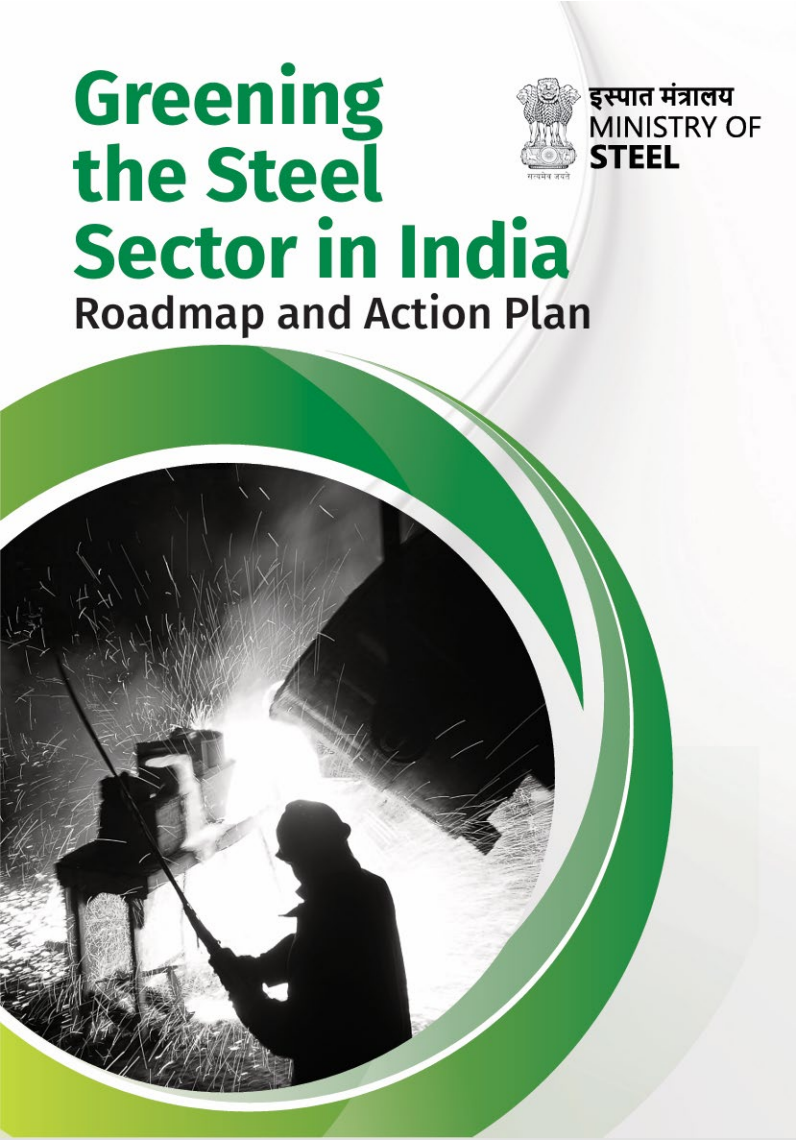


Technology process	reductant	CO2 emission intensity (tCO_2/tcs, India)
BF-BOF	Coke	**2.46**
DRI-IF	Coal	**2.30**
DRI-EAF	Coal	**2.51**
	NG	**1.91**

- India's unique coal-based DRI-IF/EAF route and the traditional BF-BOF route are the two main technical routes for steel production in this country.
- The carbon emission intensity of the Coal-DRI process in India is even higher than traditional BF-BOF route.

Case India: Review

Carbon intensity and decarbonization policies



Timeline	Objective
2030	• Reduce integrated emission intensity to 2.20 tCO_2/tcs • 45% penetration rate of renewable energy • Significantly increase the utilization of scrap steel and pellet ore • Demonstrate pilot plants using green hydrogen
2070	• Achieve net-zero emissions for the steel industry

- **India's steel industry has high carbon intensity:** 2.36 to 2.6 tCO2/tcs, compared to the global average of 1.85 to 1.91 tCO_2/tcs; Contributes ~2% to India's GDP, but ~12% of the CO2e.
- The Indian government **supports the development and promotion** of hydrogen-based steelmaking in India.
- We also need to **develop a transformation pathway** for India's steel sector decarbonization plan by 2070.

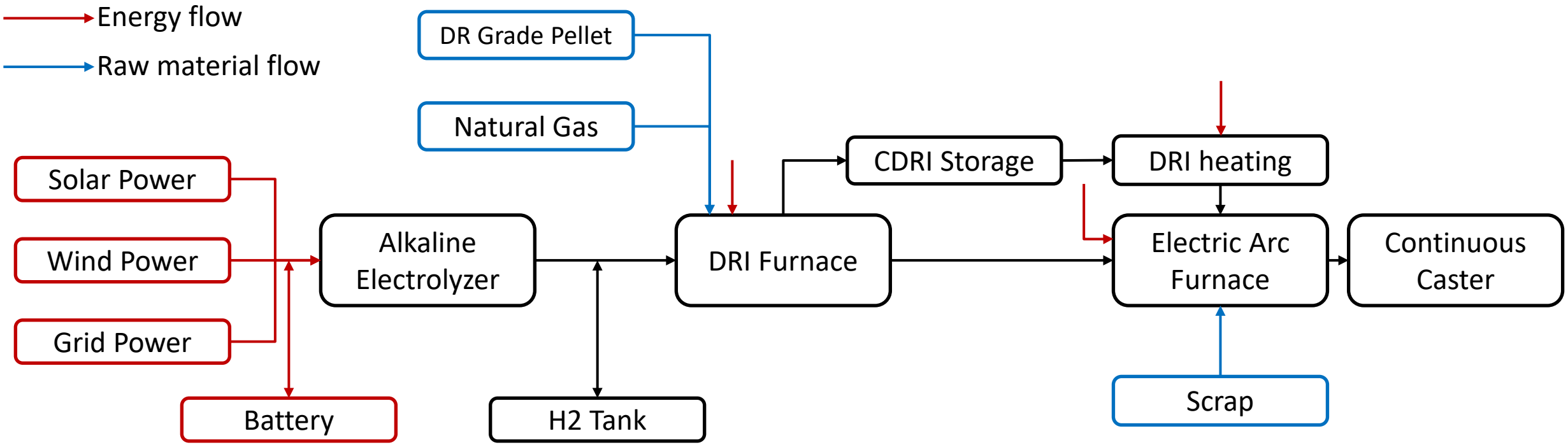
Model development

Current model: H2-DRI-EAF production scheduling model with flexibility

Hourly resolution system optimization model, **Minimizing LCOS**

Consider the **production, storage and consumption** of **electricity, hydrogen, iron and steel**

Flexibility modules: battery, H2 tank, CDRI storage



We will continue to update this model in subsequent research, and update the results in the Indian Case.

Data collection in Indian case (Minimum standard)

Data type	Data requirements	Source
Renewable output	• Hourly • Solar & Wind • Weather conditions	• Renewables.ninja • ERA5, MERRA-2
CAPEX & OPEX	• Solar, wind, battery, electrolyzer, …	• IRENA, BP • Related researches
Labor cost	• Ironmaking, Steelmaking	• Related researches
Energy / Feedstock consumption	• DRI, EAF, electrolyzer, …	• Same as China's scenario
Carbon emission intensity	• Grid power, Gas	• Same as China's scenario

- For the minimum standard requirements, we only need to select a location, find data from the corresponding sources, and then run the model to obtain the results.
- Through this way, we can obtain the H2-DRI-EAF operating mode and minimum LCOS at a certain location in India.

Database: Labor cost

- Literature review: ***Global green hydrogen-based steel opportunities surrounding high quality renewable energy and iron ore deposits, Nature Communications, 2023***
  - Calculate steel labor costs using hourly labor cost and labor intensity
  - Empirical formula: **Steelworker wage = Average wage * 1.3**
  - hourly labor cost = Steelworker wage / (1 + Tax Rate)

- **Using this method, the labor cost per ton of steel in India can be calculated.**

Table S18 – Regionally variable steelworker wages

	Average wage (before tax)	Steelworker wage (before tax)		Personal Income Tax Rate (highest)	Steelworker hourly rate, inc. tax
Source	[47]	30% increase above average [2]		[48]	
Unit	$/year	$/year	$/hr	% of income	$/hr
AU	53620	69706	36.31	45%	52.64
BR	7820	10166	5.29	28%	6.75
CA	43540	56602	29.48	33%	39.21
CL	12990	16887	8.80	40%	12.31
CN	10520	13676	7.12	45%	10.33
GN	960	1248	0.65	40%	0.91
IN	1890	2457	1.28	43%	1.83
IR	3290	4277	2.23	35%	3.01
KZ	8710	11323	5.90	10%	6.49
MX	8750	11375	5.92	35%	8.00
NZ	41480	53924	28.09	39%	39.04
PE	6000	7800	4.06	30%	5.28
RU	10740	13962	7.27	13%	8.22
ZA	6090	7917	4.12	45%	5.98
SE	54740	71162	37.06	52%	56.34
TR	9070	11791	6.14	40%	8.60
UA	3570	4641	2.42	18%	2.85
US	64650	84045	43.77	37%	59.97

Table S19 – Labour intensities

Process	Labour intensity	Source
Green H2 production	2 hours/kW installed electrolysers	Own calculations using data from Leguijt, et al. [49], considering labour demand for RE generation and electrolysis parts of value chain.
DR ironmaking	0.22 hours/t DRI	Own calculations using data from Wood, et al. [2].
EAF steelmaking	0.49 hours/t steel	

Thank You For Your Attention!